AI Internship – Day 3 Report
Training and Evaluating Machine Learning Models (Regression & Classification)

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Objective

The objective of Day 3 was to:

- Train machine learning models using Scikit-learn
- Apply Linear Regression for house price prediction
- Apply Logistic Regression for classification (Titanic survival prediction)
- Evaluate models using appropriate performance metrics
- Interpret model behavior using coefficients and feature importance

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📊 Datasets Used

1. House Prices Dataset (Regression)

- Target: `median_house_value`
- Features: numerical + categorical housing attributes

2. Titanic Dataset (Classification)

- Target: `Survived (0/1)`
- Features: passenger demographics and travel details

🧠 Concepts Learned

🔹 1. Train-Test Split

Data was divided into:

- **Training set (80%)** → model learning

- **Testing set (20%)** → model evaluation

Stratified sampling was used for Titanic to maintain class balance.

🔹 2. Linear Regression (Regression Task)

Used to predict continuous values (house prices).

Model learns a linear relationship:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

🔹 3. Ridge & Lasso Regression

Used to reduce overfitting:

- **Ridge Regression** adds L2 penalty
- **Lasso Regression** adds L1 penalty and can eliminate features

$$RSS + \lambda \sum_{j=1}^{p} \beta_j^2$$

$$RSS + \lambda \sum_{j=1}^{p} |\beta_j|$$

🔹 4. Logistic Regression (Classification Task)

Used for binary classification (survived / not survived).

Outputs probability:

- Converts probability into class labels (0 or 1)

🔹 5. Evaluation Metrics (Regression)

- **RMSE (Root Mean Squared Error):** Measures average prediction error

- **R^2 Score:** Measures how well the model explains variance in data

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🔹 6. Evaluation Metrics (Classification)

- **Accuracy:** Overall correctness of predictions
- **Confusion Matrix:** Breakdown of TP, TN, FP, FN
- **Precision, Recall, F1-score:** Performance balance
- **AUC Score:** Ability to distinguish between classes

📊 Model Performance Results

🏠 House Prices (Linear Regression)

- **RMSE:** ~70,059
- **R^2 Score:** ~0.625

Ridge Regression

- RMSE: ~70,066
- R^2: ~0.625

Lasso Regression

- RMSE: ~70,059
- R^2: ~0.625

📌 Observation:

```
Model Comparison:
Linear Regression R2: 0.6254382675296266
Ridge R2: 0.6253652564019248
Lasso R2: 0.6254370890674854
```

All models performed similarly, indicating:

- Linear relationships dominate dataset
- Regularization had minimal impact

🚢 Titanic Dataset (Logistic Regression)

```
AUC Score: 0.8532279314888009
Accuracy: 0.8268156424581006
AUC: 0.8532279314888009

Confusion Matrix:
 [[98 12]
 [19 50]]
```

- **Accuracy:** 0.8268 (~82.6%)
- **AUC Score:** 0.853

Confusion Matrix:

- Correctly predicted non-survivors: 98
- Correctly predicted survivors: 50
- Misclassifications: 31 total

📌 Observation:

- Model performs well overall
- Slight weakness in detecting all survivors (lower recall for class 1)

📈 Feature Interpretation

- Higher income and favorable location increase house prices
- Certain passenger features (e.g., gender, class) strongly influence survival probability

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🛠️ Tools & Libraries Used

- Python
- Pandas, NumPy
- Scikit-learn:
    - LinearRegression
    - LogisticRegression

- Ridge, Lasso
- Train_test_split
- Pipeline, ColumnTransformer
- Metrics (RMSE, R^2, Accuracy, F1, AUC)

- Joblib (for model saving)

🧠 Key Learnings

- Importance of separating training and testing data
- How pipelines automate preprocessing and prevent data leakage
- Difference between regression and classification problems
- Interpretation of evaluation metrics
- Impact of regularization techniques
- Real-world ML workflow using Scikit-learn

🏁 Conclusion

Day 3 focused on building and evaluating machine learning models. Linear Regression was applied to predict house prices, while Logistic Regression was used for survival prediction on the Titanic dataset. The models were evaluated using appropriate metrics, and results showed strong baseline performance. This task strengthened understanding of supervised learning, evaluation techniques, and model interpretation.